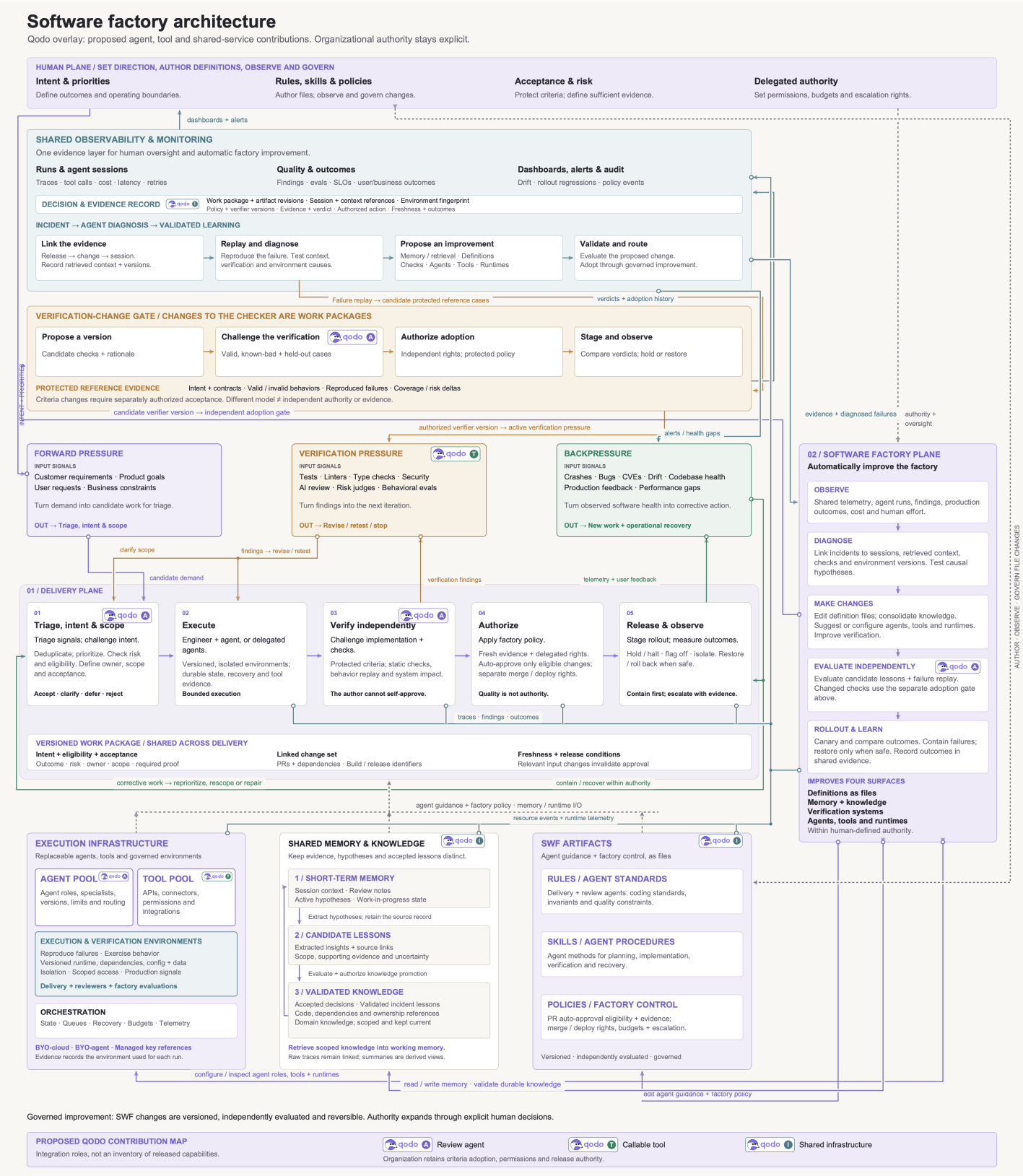
Software factory architecture | v12

A4 print edition. Page 1 is the overview; pages 2-5 enlarge the content for reading. Digital map: 3420 x 3940.



Print at actual size. Qodo badges show proposed contributions, not an inventory of released capabilities.

Human authority, evidence and learning

The organization defines the boundaries; delivery and factory improvement share the same evidence layer.

Human plane

Intent & priorities
Define outcomes and operating boundaries.

Rules, skills & policies
Author files; observe and govern changes.

Acceptance & risk
Protect criteria; define sufficient evidence.

Delegated authority
Set permissions, budgets and escalation rights.

Shared observability and monitoring

One evidence layer for human oversight and automatic factory improvement.

Runs & agent sessions
Traces · tool calls · cost · latency · retries

Quality & outcomes
Findings · evals · SLOs · user/business outcomes

Dashboards, alerts & audit
Drift · rollout regressions · policy events

Decision and evidence record [I]
Work package + artifact revisions · Session + context references · Environment fingerprint
Policy + verifier versions · Evidence + verdict · Authorized action · Freshness + outcomes

Incident -> agent diagnosis -> validated learning

Link the evidence
Release → change → session.
Record retrieved context + versions.

Replay and diagnose
Reproduce the failure. Test context, verification and environment causes.

Propose an improvement
Memory / retrieval · Definitions
Checks · Agents · Tools · Runtimes

Validate and route
Evaluate the proposed change.
Adopt through governed improvement.

Failure replay supplies candidate protected reference cases. Diagnosed failures and evidence feed the SWF improvement plane.

Verification-change gate

Propose a version
Candidate checks + rationale

Challenge the verification [A]
Valid, known-bad + held-out cases

Authorize adoption
Independent rights; protected policy

Stage and observe
Compare verdicts; hold or restore

Protected reference evidence: Intent + contracts · Valid / invalid behaviors · Reproduced failures · Coverage / risk deltas

Criteria changes require separately authorized acceptance. Different model ≠ independent authority or evidence.

Only the authorized verifier version becomes active verification pressure. Verdicts and adoption history return to shared observability.

Pressures and the delivery plane

Triage is part of the first delivery step, before execution. Authorization applies factory policy to fresh evidence.

Three sources of pressure

FORWARD PRESSURE

INPUT SIGNALS

Customer requirements · Product goals
User requests · Business constraints

Turn demand into candidate work for triage.

OUT → Triage, intent & scope

VERIFICATION PRESSURE [T]

INPUT SIGNALS

Tests · Linters · Type checks · Security
AI review · Risk judges · Behavioral evals

Turn findings into the next iteration.

OUT → Revise / retest / stop

BACKPRESSURE

INPUT SIGNALS

Crashes · Bugs · CVEs · Drift · Codebase health
Production feedback · Performance gaps

Turn observed software health into corrective action.

OUT → New work + operational recovery

Delivery flow: 01 -> 02 -> 03 -> 04 -> 05

Triage, intent & scope [A]

Triage signals; challenge intent.

Deduplicate; prioritize. Check risk and eligibility. Define owner, scope and acceptance.

Accept · clarify · defer · reject

Execute

Engineer + agent, or delegated agents.

Versioned, isolated environments; durable state, recovery and tool evidence.

Bounded execution

Verify independently [A]

Challenge implementation + checks.

Protected criteria; static checks, behavior replay and system impact.

The author cannot self-approve.

Authorize

Apply factory policy.

Fresh evidence + delegated rights. Auto-approve only eligible changes; separate merge / deploy rights.

Quality is not authority.

Release & observe

Stage rollout; measure outcomes.

Hold / halt · flag off · isolate. Restore / roll back when safe.

Contain first; escalate with evidence.

Versioned work package, shared across delivery

Intent + eligibility + acceptance

Outcome · risk · owner · scope · required proof

Linked change set

PRs + dependencies · Build / release identifiers

Freshness + release conditions

Relevant input changes invalidate approval

Forward pressure supplies candidate demand. Verification findings return work to execution or scope clarification. Production feedback creates corrective work and can trigger containment or recovery within delegated authority.

Relevant input changes invalidate approval. Quality evidence does not itself grant authority to merge or deploy.

The shared resources

Execution infrastructure is on the left, memory in the middle, and SWF artifacts beside the factory-improvement plane.

Execution infrastructure

AGENT POOL [A]

Agent roles, specialists,
versions, limits and routing

TOOL POOL [T]

APIs, connectors,
permissions and
integrations

EXECUTION & VERIFICATION ENVIRONMENTS

Reproduce failures · Exercise behavior
Versioned runtime, dependencies, config + data
Isolation · Scoped access · Production signals

Delivery + reviewers + factory evaluations

ORCHESTRATION

State · Queues · Recovery · Budgets · Telemetry

BYO-cloud · BYO-agent · Managed key references

Evidence records the environment used for each run.

Shared memory and knowledge [I]

1 / SHORT-TERM MEMORY

Session context · Review notes
Active hypotheses · Work-in-progress state

2 / CANDIDATE LESSONS

Extracted insights + source links
Scope, supporting evidence and uncertainty

3 / VALIDATED KNOWLEDGE

Accepted decisions · Validated incident lessons
Code, dependencies and ownership references
Domain knowledge; scoped and kept current

Short-term memory -> extract hypotheses with source links -> candidate lessons -> evaluate and authorize promotion -> validated knowledge. Retrieve scoped knowledge into working memory.

Raw traces remain linked; summaries are derived views.

SWF artifacts [I]

RULES / AGENT STANDARDS

Delivery + review agents: coding standards,
invariants and quality constraints.

SKILLS / AGENT PROCEDURES

Agent methods for planning, implementation,
verification and recovery.

POLICIES / FACTORY CONTROL

PR auto-approval eligibility + evidence;
merge / deploy rights, budgets + escalation.

Versioned · independently evaluated · governed

Rules and skills guide delivery and review agents. Factory policies govern the workflow, including PR auto-approval eligibility, required evidence, merge/deploy rights, budgets and escalation. Humans author and govern these files; the SWF plane improves them within delegated authority.

The software factory improvement plane

The second job of the factory is to improve how future work gets done.

Observe -> Diagnose -> Make changes -> Evaluate -> Rollout and learn

OBSERVE

Shared telemetry, agent runs, findings, production outcomes, cost and human effort.

DIAGNOSE

Link incidents to sessions, retrieved context, checks and environment versions. Test causal hypotheses.

MAKE CHANGES

Edit definition files; consolidate knowledge. Suggest or configure agents, tools and runtimes. Improve verification.

EVALUATE INDEPENDENTLY [A]

Evaluate candidate lessons + failure replay. Changed checks use the separate adoption gate above.

ROLLOUT & LEARN

Canary and compare outcomes. Contain failures; restore only when safe. Record outcomes in shared evidence.

Four improvement surfaces

Definitions as files; memory and knowledge; verification systems; agents, tools and runtimes. Changes remain within human-defined authority.

Connections and decision rights

Human authority

Humans define intent, acceptance, risk and delegated authority. They author, observe and govern the artifact files.

Shared evidence

Delivery runs, quality outcomes and resource/runtime events feed observability. Humans and the SWF plane consume the same evidence.

Resource interfaces

Memory and execution infrastructure have bidirectional interfaces: context retrieval and updates, calls and results. Delivery consumes agent guidance and factory policy.

Governed changes

The SWF plane edits guidance and policy, validates durable knowledge, and configures or inspects agent roles, tools and runtimes. Changed verification uses the separate adoption gate.

SWF changes are versioned, independently evaluated and reversible. Authority expands through explicit human decisions.

Proposed Qodo contribution map

[A] Review agent [T] Callable tool [I] Shared infrastructure

Integration roles, not an inventory of released capabilities. The organization retains criteria adoption, permissions and release authority.